
Part I

Basics

Injury Statistics

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Yasser El-Sheikh at Rifle canyon, USA, Photo Michael Simon

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2.1 Introduction

Accurate estimates of injury burden are required to develop appropriate preventative and rehabilitative strategies. Many studies have already presented injury analysis for sport climbing, bouldering, and alpine climbing [1–22]. Unfortunately, many studies fail to differentiate between the various types of climbing disciplines. Some studies focus specifically on indoor climbing [16, 17, 23–25] or competition climbing [17, 22, 26, 27]. Concerning the specific injury incidence rate, sports medicine performs analyses of the injury risk per 1000 h of sports performance, a procedure which can be difficult in climbing, as the climbing time of a single climbing day is more difficult to analyze than the 90 min of play during a soccer game [28–30]. In other mountain sport activities such as high-altitude mountaineering, injury risks differ largely from those present in climbing [28–30]. To objectively analyze and compare injuries from different sports, a common scoring system for the grading of the injuries is essential. In general, there should be a distinction between overstrain (overuse) injuries and acute injuries or accidents when assessing whether a sport is “high-risk” or not.

Overstrain injuries are usually less severe and can generally be avoided with informed training. An examination of the injury rate for acute sport-specific injuries, especially their severity, is crucial. When considering the question of whether a sport is high-risk, overstrain injuries are to be neglected. A standardized scoring scale is essential to compare injury risk and severity.

2.2 Injury Grading and Scoring

In classic outdoor rock climbing, the historic studies of Schussmann et al. [3] and Bowie et al. [2] were the first to grade registered injuries with a scoring model (ISS-score) to investigate the connection between specific injury risk and climbing days [2] or climbing time [3]. Unfortunately, the ISS-score showed weak validity for injuries in a self-recall [31]. Other studies used the NACA score (National Advisory

Committee for Aeronautics) [29, 32–34] or other general trauma scores. The NACA score, which originated in the field of aviation rescue, is the most popular pre-hospital emergency medical score in Germany and is an essential part of the pre-hospital emergency rescue documentation [35]. In Europe, this scale is also recommended internationally for alpine trauma and its evaluation [36]. The disadvantage of the NACA score is that it is a pre-hospital score and does not consider the final patient outcome. A NACA score of 3–6 can either have a good or a fatal outcome and is therefore not appropriate [30]. An inter-study comparison both within the sport and with other sports is difficult to execute since many studies in the past used different scoring models. All of these scoring systems have proven flawed for the evaluation of mountaineering and climbing injuries [28, 34], as they do not present treatment outcome, give sport-specific references such as climbing grades, or assess the climbing risk. Therefore, the Medical Commission of the UIAA (Union Internationale des Associations d’Alpinisme) developed its own scoring model which allows a sport-specific injury grading [30] (Table 2.1). This score was recommended for climbing and mountaineering injury and accident analysis to guarantee an inter-study comparability. In an extension of this classification, the IFSC (International Federation of Sport Climbing) presented their own scoring model analyzing the World Cup injuries in 2012 [17]. For a fatality analysis in climbing and mountaineering accidents, the UIAA Medical Commission also presented a risk classification [30] (Table 2.2). This is comparable to the British “E” scale of climbing risk, which considers not only the physical demand of the climb but also its danger, exposure, and rock quality.

In order to compare the risk of injury between climbing sports and other sports, studies should calculate an injury risk per 1000 h of sport-specific performance (time-related injury risk, morbidity rate) as this measure is increasingly used in scientific analysis. As sport-performance time analysis in climbing presents some difficulty, the UIAA Medical Commission (UIAA MedCom) proposed the following:

Table 2.1 UIAA MedCom Score 2011 (Schöffl et al. [30])

Injury and Illness Severity Classification (IIC) – UIAA MedCom Score	
0. No injury or illness	
1.	Mild injury or illness, no medical intervention necessary, self-therapy (e.g., bruises, contusions, strains)
2.	Moderate severe injury or illness, not life threatening, prolonged conservative or minor surgery, outpatient therapy, doctor attendance within a short time frame (days), injury-related work absence, heals without permanent damage (e.g., undisplaced fractures, tendon ruptures, pulley ruptures, dislocations, meniscal tear, minor frostbite)
3.	Major injury or illness, not life threatening, hospitalization, surgical intervention necessary, immediate doctor attendance necessary, injury-related work absence, heals with or without permanent damage (e.g., dislocated joint, fractures, vertebral fractures, cerebral injuries, frostbite with amputations)
4.	Acute mortal danger, polytrauma, immediate prehospital doctor or experienced trauma paramedic attendance if possible, acute surgical intervention, outcome: alive with permanent damage
5.	Acute mortal danger, polytrauma, immediate prehospital doctor or experienced trauma paramedic attendance if possible, acute surgical intervention, outcome: death
6.	Immediate death

Table 2.2 Fatality Risk Classification (FRC) (UIAA MedCom (Schöffl et al. [30]))

I	Fatalities are technically possible but very rare. No objective danger, e.g., indoor climbing
II	Few objective dangers, fatalities rare, falls are not very dangerous, risk is mostly calculable, e.g., sport climbing, low elevation, and technically easy peaks
III	High objective danger, risk is difficult to calculate, falls lead frequently to injuries, fatalities more frequent, e.g., traditional climbing, high Himalayan (7000–8000 m), or difficult peaks
IV	Extremely dangerous, falls have a high fatality rate, totally unjustified to normal mortals

“If the hours are not collected for individual rock climbing days, each day can be calculated as four hours for sport and traditional climbing, eight hours for alpine climbing, two hours for indoor climbing, six hours for ice climbing and sixteen hours for an expedition day. The sixteen-hour expedition day estimate is based on twelve hours of climbing time, four hours of camp activity and eight of hours sleep

Table 2.3 Functional outcome after finger injuries in climbers (Lutter et al. [22])

Excellent: free range of motion, no objective strength deficit, normal motion pattern
Good: extension or flexion deficit of the PIP joint up to 10°, minor strength deficit, normal motion pattern
Satisfactory: extension or flexion deficit of the PIP joint up to 20°, clinical strength deficit, minor irritation of motion pattern
Fair: extension or flexion deficit of the PIP joint of more than 20°, major strength deficit, major irritation of motion pattern

PIP proximal interphalangeal joint

This score is based on the original functional outcome score after pulley injuries (Schöffl et al. [39])

and reflects the very small risk of an injury being sustained during camp time and sleep (e.g. avalanche). All of these hours (especially for the expedition day) are estimations and have been agreed upon by the UIAA Medical Commission to enable calculations of injury risk per 1000 hours of sport-specific performance” [30].

Other concepts use a climbing intensity score (CIS) [37] or calculate athletic load. The latter combines weighted operation measures of performance including frequency of ascent, specific climbing behavior, grade of difficulty, and rate of perceived exertion (RPE) to produce an arbitrary unit [38].

Specific outcome scores have been proposed and implemented for the climbing-specific outcome in analytic studies of injuries and therapeutic concepts. Schöffl et al. [39] showed a functional as well as a sport-specific outcome score in the analysis of finger pulley injuries in climbers. This scoring system was later extended to include all finger injuries [40] (Tables 2.3 and 2.4).

2.3 Analysis of the Climbing Subdisciplines

For further analysis, an exact differentiation in between the various climbing subdisciplines is necessary. The geographic origin of the data also must be considered in order to account for the different situations present in various geographic regions and climbing areas. Serious climbing injuries may have a worse outcome in

Table 2.4 Sport-specific outcome score after finger injury in climber (Lutter et al. [22])

Excellent: full load capacity of the former injured finger after 12 months (with or without tape), no subjective strength deficit of the former injured finger, regain of full climbing ability/pre-injury climbing level, no pain

Good: full load capacity of the former injured finger after 12 months (with tape), subjectively minor strength deficit of the former injured finger, regain of full climbing ability/pre-injury climbing level, minor pain

Satisfactory: minor restricted load capacity of the former injured finger after 12 months (with tape), subjectively strength deficit of the former injured finger, regain of full climbing ability/pre-injury UIAA climbing level minus one UIAA grade, minor pain

Fair: major restricted load capacity of the former injured finger after 12 months (with tape), strength deficit and restricted ability to use the former injured finger while climbing, major decrease in climbing ability and grade, frequent pain

Poor: climbing is not possible anymore

This score is based on the original sport-specific outcome score after pulley injuries in climbers (Schöffl et al. [39])



Fig. 2.2 Sport climbing with low risk of objective danger. (Image credit: Michael Simon)

climbing. For completeness, water ice climbing will be also presented.

2.3.1 Alpine Climbing, Sport Climbing, and Bouldering

Unfortunately, analysis of the literature shows that few studies differentiate precisely between these disciplines [42]. Therefore, these disciplines are investigated collectively. Many studies focus on hand and finger injuries and only give little data concerning injuries [28, 29]. Looking at analyses of the most frequent injuries in climbing, the injury distribution is very much depending on the participant selection. Buzzacott et al. [8] reported on the NEISS database (US) for the years from 2008 to 2016. The NEISS database reports all emergency medical patients of the US NEISS hospitals, which then can be analyzed through keyword tracing (e.g., “climbing”) [43]. Buzzacott et al. [8] found that 47% of all injuries are to the lower extremities and are mostly fractures and sprains. This finding is similar to the prior years of 1990–2007 in the NEISS database as shown by Nelson et al. [43]. These data collections do not account for minor and very sport-specific injuries (e.g., finger pulley injuries or finger joint capsulitis), as affected patients normally do not present to an emergency department, but will be seen in specialized



Fig. 2.1 Alpine climbing with objective danger (rock fall, weather conditions) and potential long falls due to mostly clean protection with nuts and friends. (Credit: Archive Adidas Terrex)

a developing country or a remote climbing location when compared to an injury in the Alps or other areas with a well-developed mountain rescue infrastructure and optimal medical care [28, 29, 41]. We are going to distinguish in the following as much as possible between alpine climbing (Fig. 2.1), sport climbing (Fig. 2.2), bouldering, indoor climbing, and competition

sport-orthopedic centers. Looking at our database of climbing-specific patients over various time frames, we mostly see overstrain injuries to the upper extremities, mostly to the hand, fingers, and shoulders [15, 22, 44]. This corresponds with the findings of many other studies which state that 58–67% of the injuries are to the upper extremities, examining climbers in cross-sectional and longitudinal studies [4–6, 9, 10, 12, 37, 44, 45]. Nevertheless, our data also have a selection bias, as we see many climbers to provide a second opinion on very sport-specific injuries (e.g., pulley injuries), whereas acute traumas (e.g., ankle fracture) receive primary treatment close to the geographic location of the accident or the patient's residence. In conclusion, it can be said that most acute traumatic injuries (fractures, sprains) occur to the lower extremities, while most sport-specific injuries occur to the upper extremities and are often chronic and related to overstrain. Sport climbing injuries occur most frequently during lead climbing [1, 2, 4, 7, 9, 22, 28, 29]. The most frequent cause of injury is a fall [1, 2, 4, 7, 9, 28, 29]. Looking at the subdisciplines, alpine climbing injuries are more likely to be to the lower extremities, usually caused by falls. In sport climbing, most injuries occur to the upper extremities, usually while performing a strenuous move [4, 6, 9, 10, 12, 46]. Bouldering shows an almost even injury distribution between fall injuries (sprains and strains) and overstrain. Josephsen et al. [47] found no difference between indoor and outdoor bouldering. Recently, an the incidence of bouldering injuries has increased [14]. The overall injury incidence is minor and was recently reported by Jones et al. [9] through a meta-analysis of eight studies with 2.7 ± 4.5 per 1000 h of sport. These injury rates are lower compared to those of motorcycling with 13.5 per 1000 h of sport [48], soccer with 31 per 1000 h [49], handball with 50 injuries per 1000 h [50], ice hockey with 83 injuries per 1000 h [51], or rugby with 286 injuries per 1000 h of sport [52].

Concerning the injury severity, rock climbing studies show mostly minor injury severities (NACA-score 1 and 2) with a fatality rate of 0–28% [1–3, 5, 6, 9, 44, 47, 53]. The vast span between these numbers must be further evaluated

through ongoing studies and may reflect the bias of injuries recorded in the study. For alpine climbing, a 1988 study from Bowie et al. [2] reported a rate of 13 fatalities out of 220 climbing accidents. Eleven climbers died on site and the overall “case-fatality-rate” was 6%. It needs to be stated that these data are over 30 years old and the safety equipment has largely improved since then, decreasing the injury risk [41]. Gatterer et al. [54] reported that the mortality during hiking, trekking, and biking in the mountains was lower compared to that during paragliding or during rock, ice, or high-altitude climbing. Traumatic deaths were more common in activities primarily performed by young adults.

Most studies are performed retrospectively, which frequently causes subsequent systematic errors in the fatality analyses [41]. Historically, older studies show higher grades of injury severity and higher fatality rates [1–3]; the only prospective study on bouldering showed no fatalities at all [47]. These comparisons are hard to draw and have a high bias as the older studies examine high-altitude alpine terrain with high external and objective danger (e.g., weather, temperature, rock fall), whereas Josephsen [47] investigated bouldering, with negligible external dangers. He also found no difference in between indoor and outdoor bouldering, a finding which is also reported by Gerdes et al. [6]. Climbing frequency and climbing level correlates positively with the incidence of overstrain injuries in many studies [9, 15, 19, 23, 53, 55]. Only a few studies did not show this correlation [56]. In summary, Schussmann [3] had already concluded in 1990 that rock climbing has a lower injury risk than football and horse riding, with the obvious difference that the latter sports rarely result in fatalities, although this is arguable for horse riding [57].

2.3.2 Indoor and Competition Climbing

Several studies explored injuries and injury rates in both indoor climbing and indoor competition climbing. Wright et al. [23] evaluated the frequency of overuse injuries during the 1999 indoor

World Cup Championships ($n = 295$), in which 44% of the respondents had sustained an overuse injury and 19% had sustained an overuse injury at more than one site. Wright et al. [23] found an independent correlation to increased injuries ($p < 0.01$) when climbing harder routes, bouldering or leading versus top rope climbing, and climbing for over 10 years. Multivariate analysis removed the effect of sex as an independent predictor.

Jones et al. [55] similarly found increased numbers of overuse injuries or injuries caused by strenuous moves and fewer from fall-related injuries than in traditional and outdoor sport climbing [1, 2, 4, 7]. Two large-scale studies [24, 25] analyzed indoor climbing injuries. Limb's survey reported 55 accidents and no fatalities from 1.021 million climbing wall visits [25]. Schöffl and Winkelmann prospectively surveyed 25,163 registrants at ten climbing walls [24]. Only four significant injuries (NACA 3) were found, and no fatalities occurred; the injury risk per visit was 0.016% or 0.079 injuries per 1000 h of performance [24]. A higher injury risk rate of 3.1 per 1000 h was found at the 2005 World Championships [26], during which eighteen acute medical problems were treated, including 13 cases of skin bruises. The IFSC Medical Commission evaluates all of its competitions and has published the results of the 2012 World Cups [17]. Overall, they counted 1362 climbing days (3405 h) of lead climbing, 1083 days (27,075 h) of bouldering, and 255 days (6375 h) of speed climbing. Only five notable injuries occurred; all were grade 2 injuries. The overall injury risk was 0.74 per 1000 h. The injury risk in lead climbing was 0.29 per 1000 h, 1.47 per 1000 h in bouldering, and zero for speed climbing (Fig. 2.3).

In summary, these indoor climbing studies demonstrated a minor injury risk and severity in comparison to traditional climbing and various other sports [24–26]. Overuse injuries were commonly reported in the upper limbs, with the fingers being mostly affected.

No study reported a fatality rate, even though fatalities do occur when climbing indoors. The cause of these rare fatalities need to be investigated in future studies to determine whether



Fig. 2.3 Elbow dislocation during a boulder competition. (Credit: DAV, German Alpine Club)

climbing misadventure or pre-existing comorbidities contribute significantly to any death [58].

Nevertheless, some of these findings may need to be re-evaluated or are in progress of change. Just recently, we reported an increase of injury severity in indoor bouldering, mostly in beginners (“newbie syndrome”) (Fig. 2.4) [14]. This increase comes parallel to the advent of many new modern bouldering gyms. Up to about 5–7 years ago, most indoor gyms were classical rope climbing gyms or bouldering gyms in a traditional way. As these traditional bouldering walls are already referred to as “old style,” we will continue to use this term. “Old style” bouldering gyms have a large number of holds in a grid style on wooden panels (Fig. 2.5). While this high density of holds allows a high variety for training in limited space, it also demands that the climber independently finds and defines the routes for training. Even though this is still considered to be an effective training method, it holds little interest for recreational climbers and beginners.

Within the last 5–7 years, a vast number of modern “new style” bouldering gyms have evolved with a completely different approach and set-up. Boulder problems are preset in various colors which reflect the grade (Fig. 2.6). These grading colors vary between the gyms. This style is rewarding even to the newest boulderers (“newbie”) because it provides an immediate sense of accomplishment. When these first “new style” gyms evolved in Europe, they not only quickly motivated boulderers from all over the area but



Fig. 2.4 High-grade spine fracture due to boulder fall from minor height (1.20 m) in a beginner climber



Fig. 2.5 “Old school” style indoor bouldering wall

also attracted former non-climbers. Currently, most of the bigger European cities have modern “new style” boulder gyms. These gyms are meeting venues for competitive athletes to train, and plenty of fun competitions take place (Fig. 2.7). Furthermore, these venues also attract leisure



Fig. 2.6 New, modern bouldering wall with color-coded preset boulder problems. (Image credit: Enrico Haase)



Fig. 2.7 Competition at an indoor bouldering wall. (Image credit: Enrico Haase)

climbers and beginners due to the aforementioned immediate success experienced when reaching the top. Thus, it is common that staff outings, manager meetings, and children’s birthday parties are being held in boulder gyms. This causes a new injury mechanism, which we called the “newbie” syndrome.

Many beginners are absolutely untrained when they start climbing. They lack the basic trained muscular system needed to stabilize the body, especially the skeletomuscular system. Due to this, a comparably minor fall can already cause major injuries. Additionally, the beginners’ equilibrium, body perception, and control are less developed, resulting in a lack of control of the body position during falls and while landing. We are seeing an increased number of patients with higher-graded injuries from falls in indoor bouldering gyms. While falls in classic old-school

boulder gyms are mostly onto the back and can be practiced, beginners are now falling from greater heights in an uncontrolled way, causing serious damage. Many of the athletes presenting with these injuries were new to the sport and had been climbing for less than 1 year [14].

In addition to the advent of boulder competitions, the route setting recently changed to include more spectacular and acrobatic movements. The new boulder problems are a combination of classical climbing and bouldering with elements adapted from parkour sports or acrobatics. This evolution of climbing style demands different skills from the competition climbers compared to the climbers of the 1990s. Most climbing moves are dynamic, very three-dimensional, and heel hooks are frequently essential [59]. Reportedly, heel hooks have already caused an increase in knee and hip injuries in climbers [59, 60]. The increase in dynamic and spectacular moves, many of them encompassing larger heights, is subsequently increasing the injury rate and severity, as falls are more unpredictable and more difficult to control [14, 60]. Concerning injury prevention in the future, route setting, fall training for climbers, and compensatory training in beginners should be considered [22, 61].

A recent study by Meyers et al. [62] analyzed speed climbing training in adolescents and found that training regularly at practices on the speed wall was associated with a self-reported history of finger growth plate injuries among elite youth competition climbers. These preliminary data on speed climbing injuries need further investigation, as only few prior studies reported about speed climbing [17, 26].

To further evaluate these trends, especially since climbing's inclusion in the Tokyo Olympic program of 2021, we performed a single-center (Sportsmedicine Bamberg) injury surveillance in 436 climbing patients with a total number of 633 independent climbing-related injuries or complaints over the recent years of 2017–2018 [22]. It was found out that 77.1% of the injuries affected the upper extremities, 17.7% the lower extremities, and 5.2% other body regions. Injury severity was overall low. The most frequent injuries overall

were finger pulley injuries (12.3%) and finger tenosynovitis (10.6%). Acute injuries were detected in 43.9% of all injuries and chronic overuse injuries in 56.1%. Bouldering accidents were the leading cause of acute injuries (60.4%). In comparison to our two prior study populations (1998–2001 and 2009–2012), we found (1) an overall decrease in upper extremity injuries, (2) an increase in lower extremity injuries, (3) a constant decrease of finger pulley injuries and epicondylitis, (4) an increase in knee injuries and shoulder dislocations, (5) an increase in adolescents' finger growth plate injuries, and (6) bouldering as the leading injury cause [22] (see Tables 2.5 and 2.6).

Table 2.5 shows the current trends within the ten most frequent climbing injuries and their dynamics over three different time spans in our database of climbing patients presented to our sports-medical center (Sportsmedicine Bamberg) [63].

Our current book focuses on sport climbing; nevertheless, a large number of sport climbers also practice other climbing disciplines, e.g., ice climbing. Insights and data on ice climbing epidemiology are also presented due to the overlap in training and outdoor execution of these disciplines (e.g., alpine climbing and ice climbing).

2.3.3 Ice Climbing

Ice climbing is experiencing an increased level of interest in the climbing community, especially in the form of water ice climbing on frozen waterfalls, glacier formations (ice caves), and also artificial outdoor ice climbing areas (e.g., Ouray Canyon, USA) (Fig. 2.8). Several years ago, Schöffl et al. had already evaluated 88 ice climbers with a retrospective questionnaire on accidents, injuries, risk management, and climbing frequency [34]. Sport activity time (ice climbing hours) was recorded, and the injury severity was analyzed using the NACA score. Mosimann [33] reported about 46 ice climbers who were rescued by the Swiss mountain rescue (Rega) after accidents. Runer et al. [64] evaluated 70 ice climbers prospectively over 1 year, and Maskovskiy et al. [65] analyzed the ice climbing demonstration competitions at the winter Olympics in Sochi.

Table 2.5 Distribution of diagnoses in our climbing clinic (ten most frequent injuries) (Lutter et al. [22])

Injuries 2017–2018 (n = 633)	n	%	Injuries 2009–2012 (n = 911)	n	%	Injuries 1998–2001 (n = 604)	n	%
Pulley injury (finger)	78	12.3	Pulley injury (finger)	140	15.4	Pulley injury (finger)	122	20.2
Tenosynovitis (finger)	67	10.6	Capsulitis (finger)	87	9.5	Epicondylitis	51	8.4
Capsulitis (finger)	49	7.7	Tenosynovitis (finger)	80	8.8	Tenosynovitis (finger)	42	7.0
Knee injury	45	7.1	SLAP tear	51	5.6	Strain finger joint capsule	37	6.1
SLAP tear (shoulder)	37	5.8	Epicondylitis	50	5.5	Skin abrasions	34	5.6
Impingement (shoulder)	34	5.4	Impingement (shoulder)	40	4.4	Back problems	24	4.0
Wrist strain	22	3.5	Strain finger flexor tendon	36	4.0	Knee injuries	14	2.3
Epicondylitis	21	3.3	Dupuytren disease	30	3.3	Fractures	14	2.3
Growth plate injuries (finger)	19	3.0	Strain finger joint capsule	25	2.7	Capsulitis (finger)	13	2.2
Spinal injuries	18	2.8	Ganglion finger flexor tendon	19	2.1	Ganglion finger flexor tendon	11	1.8

Table 2.6 Injury distribution according to body area as presented previously (data of trunk, spine, and pelvis merged). Values are n (%) (Lutter et al. [22])

Body area	2017–2018 (n = 633)	2009–2012 (n = 911)	1998–2001 (n = 604)
Finger	261 (41.2)	474 (52)	247 (41)
Shoulder	128 (20.2)	157 (17.2)	30 (5)
Hand	49 (7.7)	119 (13.1)	47 (7.8)
Forearm and elbow	49 (7.7)	83 (9.1)	81 (13.4)
Lower leg/foot	67 (10.6)	35 (3.8)	55 (9.1)
Knee	45 (7.1)	19 (2.1)	22 (3.6)
Trunk, spine, pelvis	34 (5.4)	21 (2.3)	43 (7.1)
Other	–	3 (0.3)	–

All of these studies found mostly minor graded injuries. Schöffl et al. [34] found that most of the acute injuries (61.3%) occurred while lead climbing and 23.8% while climbing second, and the rest was rare (6.3% belaying, 3.8% on return, and 2.5% on approach, other 2.5%). Most of the acute injuries (73.4%) happened in a water-fall, few in glacier ice walls (11.4%) and on arti-



Fig. 2.8 Water ice climbing with a high risk of objective danger (rock and ice fall, avalanche risk, changing weather conditions, etc.)

ficial ice walls (2.5%). Climber fall-related acute injuries amounted for 10.5%. Injuries were mainly open wounds (55.2%) and hematomas (21.9%), 71% were NACA 1, and no injury scored above NACA 3. The injury incidence was 4.07 per 1000 h for NACA 1–3 with 2.87 per 1000 h in NACA 1 and none in NACA 4–7. Overstrain injuries were mostly onto the shoulders and had an incidence of 0.77 per 1000 h. The injury risk correlated significantly with the body mass index ($p < 0.05$). Overstrain injuries correlated highly with the number of hours spent training ($p < 0.01$), the ice climbing level (grade) ($p < 0.01$), and the willingness to take risks in lead ($p < 0.01$).

Mosimann [33], reporting on 46 ice climbers rescued by the Swiss mountain rescue, found that 31% had no injury (NACA 0), 42% had NACA 2–3 injuries, 8% had NACA 4, 6% had NACA 5, and 13% (six climbers) had a fatal injury (NACA 7). The most frequent cause of an injury was a fall (55%), though falls caused no fatal injuries. The percentile death risk (fatality rate), which the author defined as the percentile portion of deaths in reference to the sum of all known emergencies, was reported as 13% for ice climbing. The author claimed that the fatality risk was higher for ice climbing than in mountaineering (8%), ski mountaineering (7.5%), and rock climbing (4%) but gave no reference for these data. Runer et al. [64] found an overall higher injury rate of 9.8 injuries per 1000 h, with this risk being the highest in the group of intermediate ice climbers. It was lower for beginners and elite ice climbers. The most frequent injuries were skin damage, contusions, and sprains. During the Winter Olympics in Sochi (16 days), more than 2500 beginners and 53 professional athletes climbed at a 17-m-high vertical artificial ice wall [65]. Injury incidence was overall low (0.83 per 1000 h). Again, the most frequent injuries which occurred were mostly minor, such as sprains, strains, and contusions.

While all of these studies show mostly minor graded trauma, serious injuries and fatal accidents can and do happen.

The general death numbers in ice climbing can be analyzed through the injury and fatality

reports of the various Alpine Clubs [34]. The Canadian [66] and the American Alpine Club [67] have statistically recorded and analyzed all mountain accidents since 1951. In the USA, up to the year 2005, there were 6111 incidents with a total of 1373 (12%) fatalities [67]. Two hundred fifty-four (4%) of the accidents happened in ice, though no further evaluation of the ice climbing injuries was given. Nevertheless, if 4% of all injuries occur during ice climbing, then 4% of the deaths can be assumed to be related to ice climbing. This calculation yields a total of 55 fatally ice climbing injuries in 54 years, an average of one ice climbing fatality per year within the USA. The numbers for Canada are similar [66]. In 30 years, 92 mountaineers were injured while ice climbing, 30 fatally. Overall, the major ice climbing countries, Switzerland (up to 2006) and Canada (1951–2004), report approximately one death per year [33, 66]. Nevertheless, these number have increased in recent winters [34]. This is probably due to the fact that ice climbing itself has become much more popular.

2.4 Is Climbing a High-Risk Sport?

Meyers encyclopedia [68] defines extreme sports as the performance of exceptional sport disciplines in which the athlete deals with high mental and physical stress. If the sport contains an objective or subjective sensed risk of damage to health or life, it is considered a high-risk sport. Meyer's definition is accurate, but it does not define any real risk from the athlete's perspective. i.e., an experienced and highly skilled athlete is more likely to take and successfully manage higher perceived sporting risks compared to a novice [28]. Kajtna and Tusak [69] define high risk sports as any sport in which one has to accept the possibility of severe injury or death as an inherent part of the activity. In contradiction, Backx et al. [70] characterized high-risk sports as those performed mostly indoors with a high jump or contact rate, as present in volleyball or basketball.

Some authors substitute the terms “high-risk sport” and “extreme sports” or use these terms

interchangeably. Young [71] included climbing under the term “extreme sports,” together with inline skating, snowboarding, mountain biking, and others. Young stated that the category of extreme sports was fluid and the definition was inexact [71]. In support of Young’s view, it is not known what the selection criteria were for an “extreme sport” to be included in the popular “X Games” media event. Climbing was repeatedly represented in this annual event.

Sport disciplines that are performed by a large population are subjectively considered harmless [72]. Therefore, more mainstream sports such as soccer, handball, or rugby are not perceived or characterized as high-risk sports, even though they have a much larger risk of injury compared to climbing [28]. Kite surfing reported a modest injury rate (7 per 1000 h) in a 6-month prospective study ($n = 235$) [73]. However, the injury incidence and severity rate was high and even recorded a fatality (124 injuries, 11 severe injuries, and 1 fatal injury) [73].

When assessing whether climbing should be considered a high-risk sport, it is obvious that each climbing subdiscipline is associated with different levels and types of risk of injury and fatality [28]. When climbing outdoors, there may be many objective dangers and physical hazards, for example, variable rock and ice quality, extreme weather conditions, weapon-like equipment (ice climbing), difficult approaches, and high mental and physical stress. In mountaineering, additional environmental factors can sometimes directly influence injuries and fatalities. These factors can include but are not limited to avalanches, crevasses, and altitude-induced illnesses with neurological dysfunction. However, these situations can still be avoided or sometimes successfully managed (i.e., weather forecasts, training in alpine climbing rescue skills, obtaining knowledge of local terrain, climbing permits, acclimatization and awareness of altitude-induced illnesses, access to helicopter mountain rescue, etc.). In contrast, the objective and external dangers are greatly reduced in indoor and sport climbing, but there is still a very low risk of a fatal injury [28].

The vast majority of climbers manage the risks with their climbing experience and skills, thereby avoiding serious injuries or fatality. Sport climbing, bouldering, and indoor climbing, including competitions, cannot be considered as high-risk sporting activities. Nevertheless, the risk of a possible fatal injury remains. Alpine, ice, and expedition climbing are all included in the broader definition of high-risk sports.

Nevertheless, based on various non-coherent definitions, many sport disciplines fulfill some criteria of a high-risk sport, even if these are often not proven through scientific numbers [41]. It is the task of sports medical science to further analyze this, which requires exact definitions of high-risk sports and injury scores which need to be developed and evaluated. For inter-study comparisons, risk evaluations per 1000 h as well as the implementation of common injury score are important tools to enable a more coherent comparison. For studies on climbing, we propose the use of the UIAA-MedCom score [30] or the IFSC MedCom score [17]).

2.5 Conclusion

In contrast to the general opinion, climbing sports show a minor injury risk and severity in comparison to many other sports. Nevertheless, some climbing subdisciplines such as alpine climbing have higher graded injuries and fatalities. The safest climbing activities are indoor, competition, sport climbing, and bouldering. Recently, there has been an increase in the reported number of injuries in indoor bouldering. This is based on the very dynamic movements of the sport and the vast increase in numbers of new beginner climbers.

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